

Total No. of Questions : 10]

SEAT No. :

P2912

[Total No. of Pages : 3

[5669]-501

T.E.(Civil)

HYDROLOGY AND WATER RESOURCES ENGINEERING

(2015 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 Or Q.No.2 Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8 and Q.No.9 or Q.No.10.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a) How hydrology is interdisciplinary science. Explain in detail. [6]
b) Write a note on drip irrigation with neat sketch state advantages and disadvantages of drip irrigation [4]

OR

- Q2) a) For a drainage basin of 600 km², isohyets drawn for a storm gave the following data [5]

Isohyetal (Interval) (cm)	15-12	12-9	9-6	6-3	3-1
Inter-isohyetal Area (Km ²)	92	128	120	175	85

Estimate the average depth of precipitation over the catchment.

- b) An irrigable canal has gross command area of 60,000 hectares out of which 70% is irrigable. The intensity of kharif and rabbi season is 25% and 50% respectively. Find discharge required at the end of canal if duty at its head is 850 hectares/cumec for kharif season and 1,800 hectares/cumec for rabbi season. [5]

Time factor = 0.9

Capacity factor = 0.8

- Q3) a) State and explain any two methods of revenue collection from farmers.[4]
b) Derive an expression for discharge from a well fully penetrating confined aquifer. [6]

OR

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Q4) a) Derive the formula to calculate discharge of a well in a unconfined aquifer. [6]

b) Define Paleo irrigation & Kor depth. Derive relation between Duty and Delta. [4]

Q5) a) What is hydrograph? Explain all the parts of the typical hydrograph. Explain fern shaped catchment. [8]

b) Given below are the ordinates of 6-h unit hydrograph for a catchment. Calculate the ordinates of the DRH due to a excess rainfall of 3.5 cm occurring in 6 h. [10]

Time (h)	0	3	6	9	12	15	18	24	30	36	42	48	54	60	69
U. H. Ordinate (m ³ /5)	0	25	50	85	125	160	185	160	110	60	36	25	16	8	0

OR

Q6) a) What is S-curve hydrograph? Explain its construction with sketch. [8]

b) In a 10 hr storm rainfall depths occurred over a the catchment are [10]

Hour	1	2	3	4	5	6	7	8	9	10
Depths (cm/hr)	1	1.5	5	6	10.5	8.5	9	7	1.5	1.5

Surface run off resulting from the storm is equivalent to 20 cm of depth over the catchment. Determine

- Average infiltration, and
- Average rate of infiltration.

Q7) a) Explain how will you fix the capacity of reservoir using annual inflow and outflow. [8]

b) Explain fixation of reservoir capacity using elevation capacity curve and dependable yield. [8]

OR

Q8) a) What are various reservoir losses. What are various measures to control these losses. [8]

b) What is reservoir sedimentation? What is the significance of trap efficiency? Explain with neat sketch. [8]

Q9) a) Write a note on ancient system of water distribution which still exist in North Maharashtra. [8]

b) Explain Global Water Partnership. (GWP) [8]

OR

Q10)a) What is water logging? Explain tile drain method and also state formula for spacing of tile drains. [8]

b) Draw a neat section for lift irrigation scheme and state various components of lift irrigation scheme. Explain various design steps in lift irrigation system. [8]

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